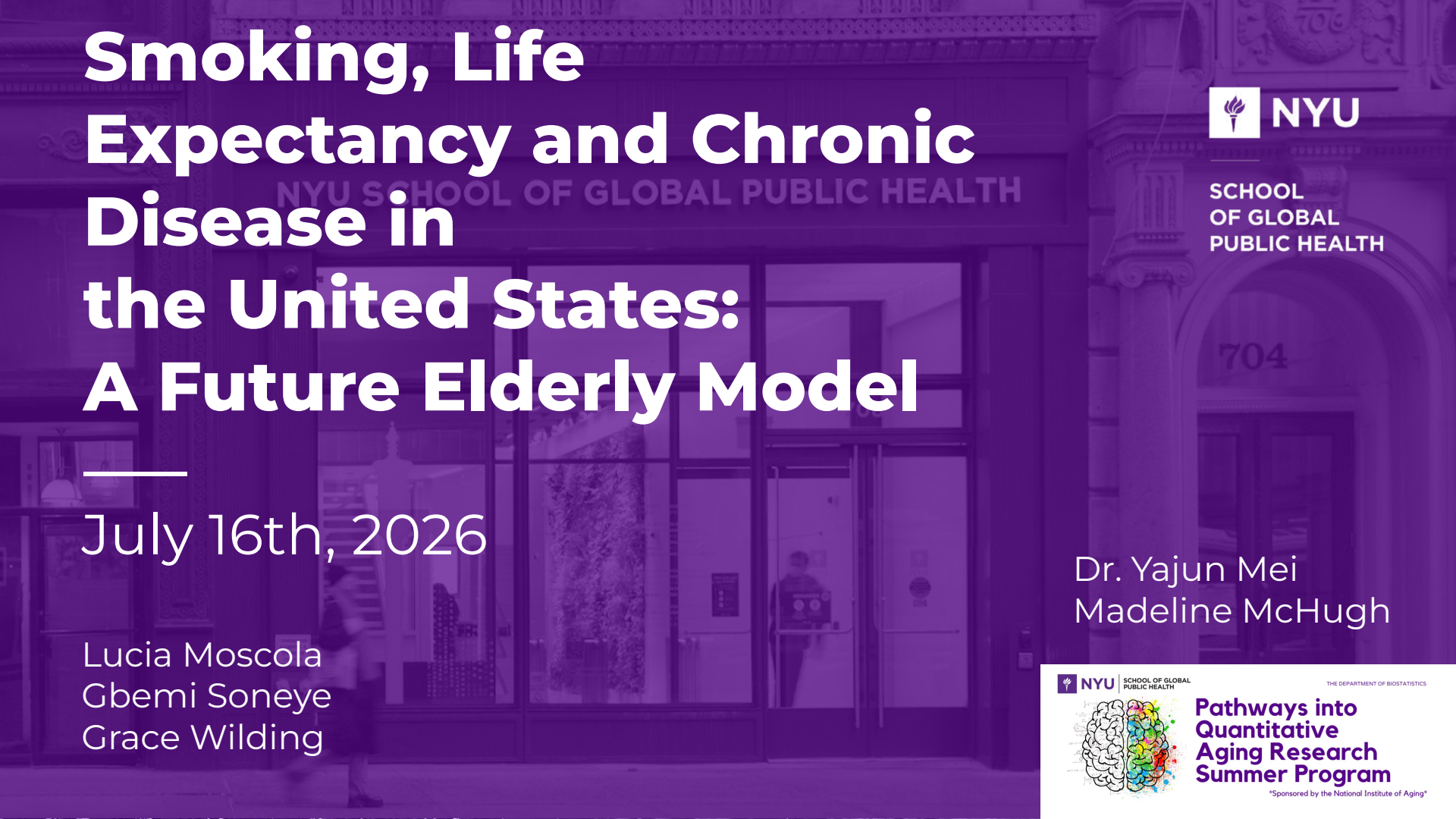
Smoking, Life Expectancy and Chronic Disease in the United States: A Future Elderly Model

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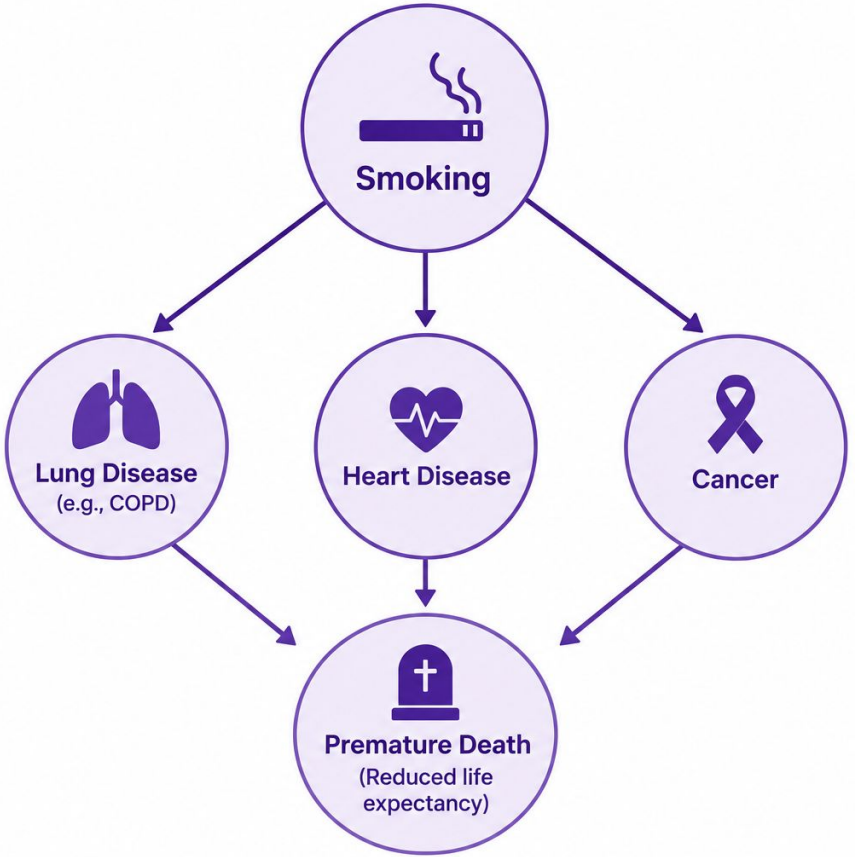
Background: Smoking in the U.S.

01

Smoking in the U.S

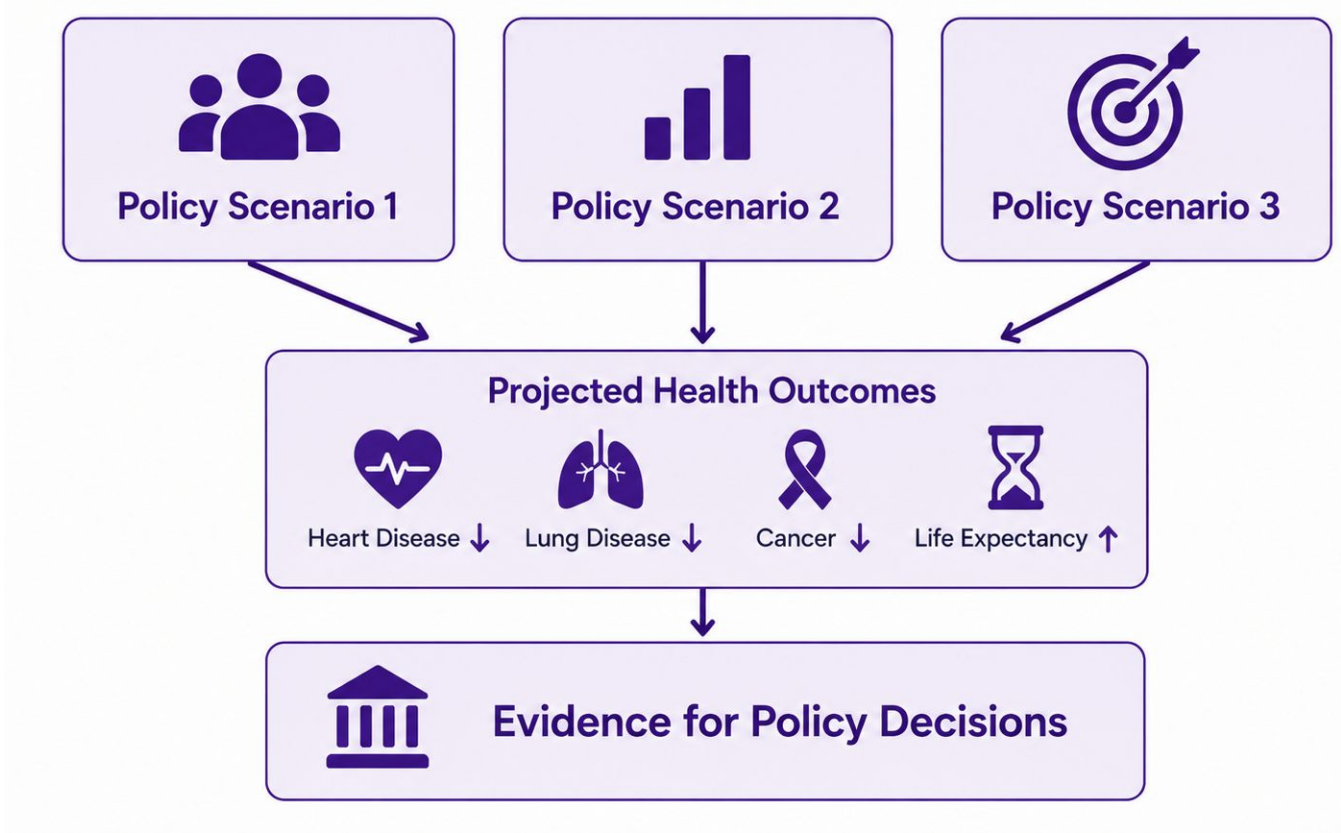
>480,000 Americans die from smoking related diseases each year.

Smoking costs the U.S. **more than $600 billion annually** in healthcare expenditures and lost productivity.



Smoking Interventions

Limited resources require evidence-based policy decisions.



Research Question

Do smoking reduction policies targeting light smokers and heavy smokers have different effects across gender?

Research Process

1. Understanding the U.S. Health and Retirement Study 1992-2022 dataset
2. Running logistic regression models to evaluate association between lifetime smoking and disease incidence
3. Making a Future Elderly Model simulation to project effects of smoking policies on expected life span and chronic disease prevalence

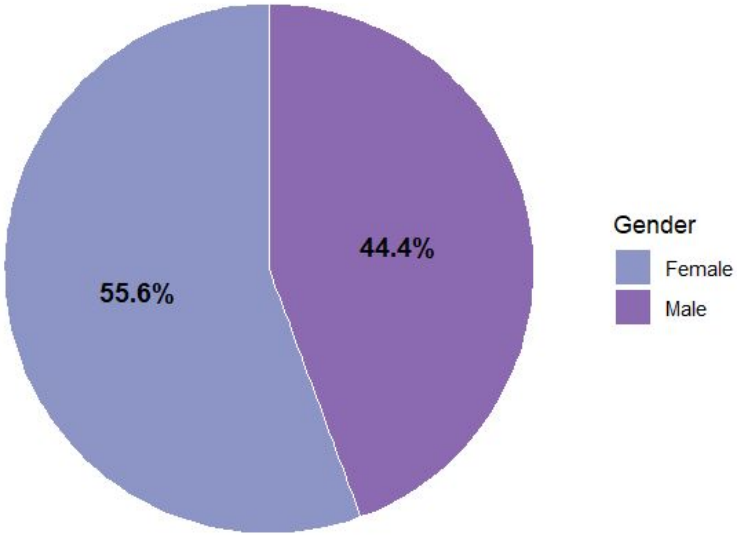
Our Data

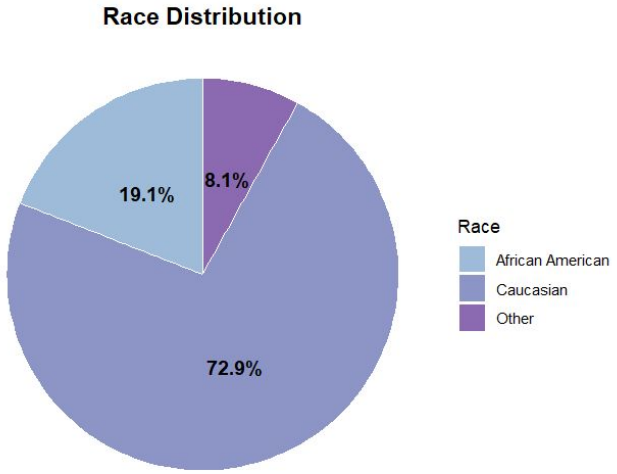
02

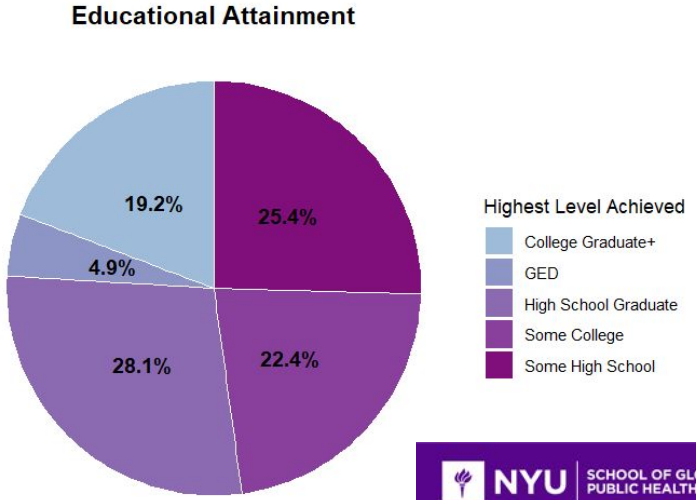
U.S. Health and Retirement Study (HRS)

- Longitudinal survey conducted by the University of Michigan, supported by the National Institute on Aging and the Social Security Administration
- ~42,000 participants across the US
- Collects demographic, health, behavioral, and social data
- Data collected in two year “waves” from 1992 to 2022
- Surveyed individuals are all 50 years or older, but some younger spouses are also included
    - We only considered data from people 50 years and older, the majority of the study

Demographics







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Smoking Data

Variables collected each wave:

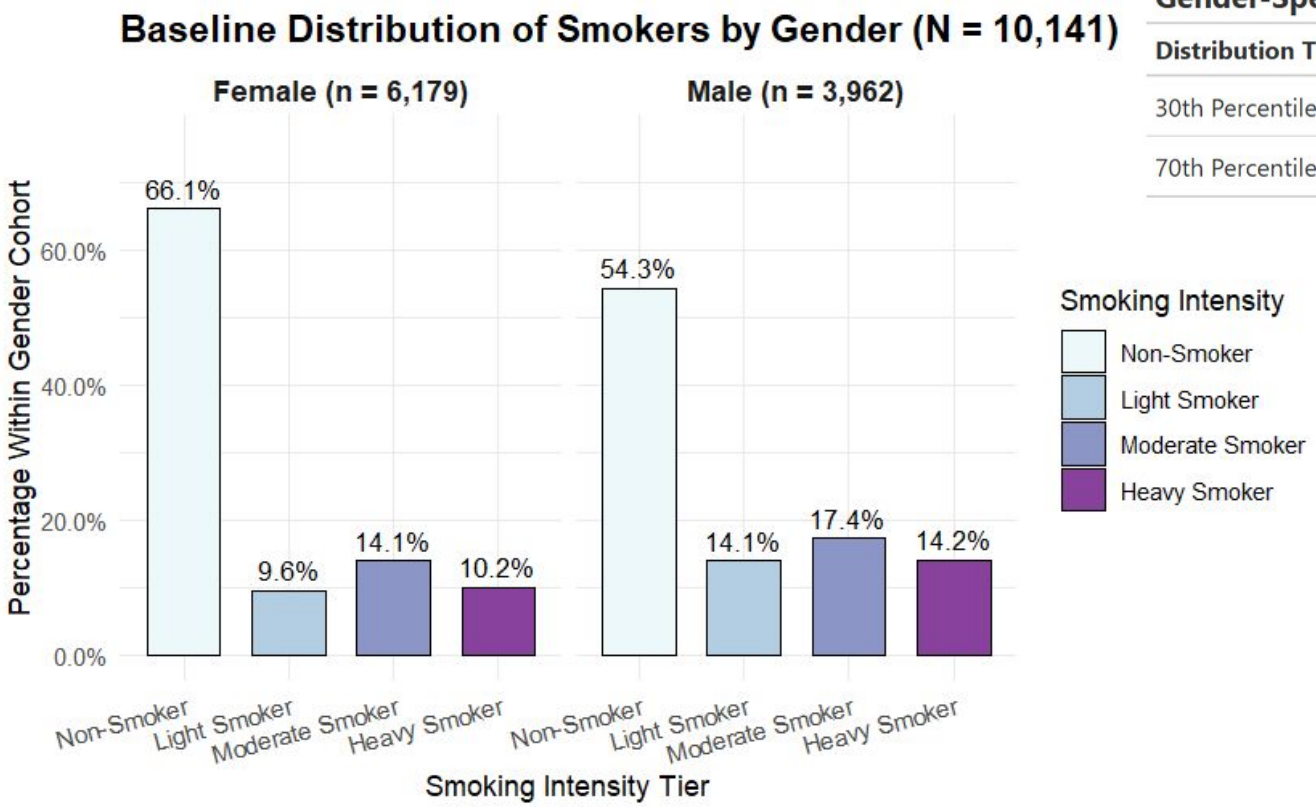
- Currently smokes (yes/no)
- Ever a smoker (yes/no)
- Age started smoking
- Age quit smoking
- Average cigarettes smoked per day

We determined:

- Pack-years to age 50
- Smoking intensity: light, medium, or heavy smoker

Pack-years: “the equivalent number of packs per day for a single year, or the equivalent number of years smoking a single pack per day” (Kim et al., 2019)

Smoking Data



Gender-Specific Smoking Intensity Cutoffs

Distribution Threshold	Male	Female
30th Percentile (Light Smoker Cutoff)	8.41	6.60
70th Percentile (Heavy Smoker Cutoff)	18.60	17.00

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Chronic Diseases in HRS

Our outcome: death and chronic disease

Population size: N = 41,369

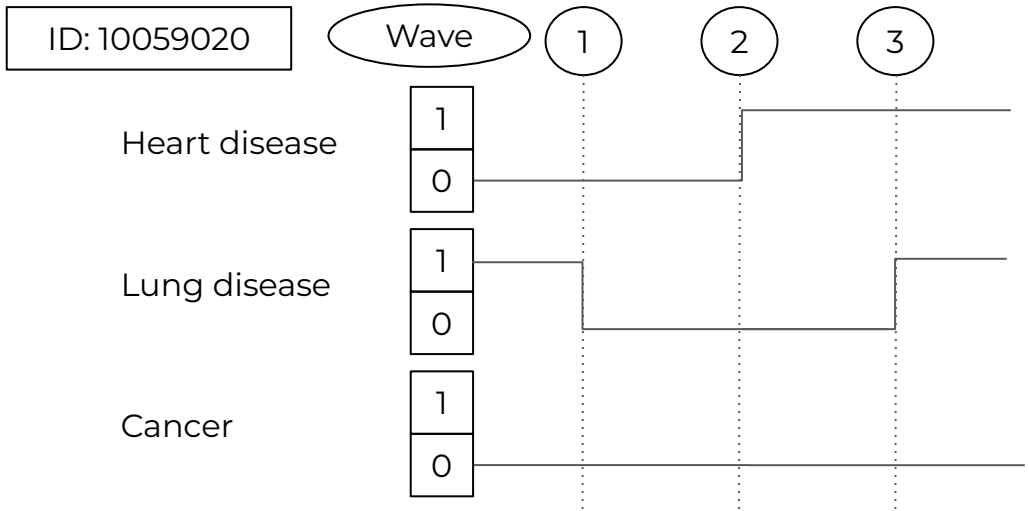
- **Death, 45.8% (n = 18,937)**
- **Lung Disease, 18.0% (n = 7,425)**
  - includes chronic bronchitis, emphysema, and asthma
- **Cancer, 21.6% (n = 8,934)**
  - includes all types of cancer (exact type was not recorded)
- **Heart Disease, 37.6% (n = 15,548)**
  - includes all types of heart disease

Methodology

03

Future Elderly Model: The Basics

- Predicting health outcomes in aging populations
- Model origin: Goldman et al., 2004
- Novel approach: modeling **individual** health transitions



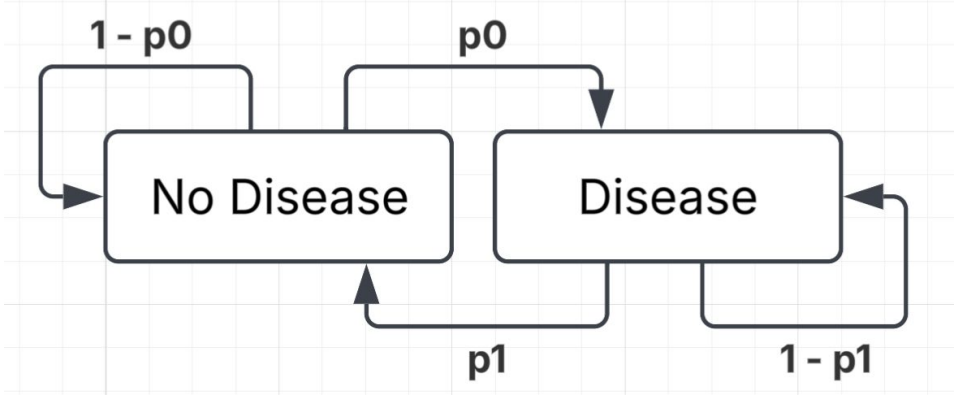
x 10,000 individuals

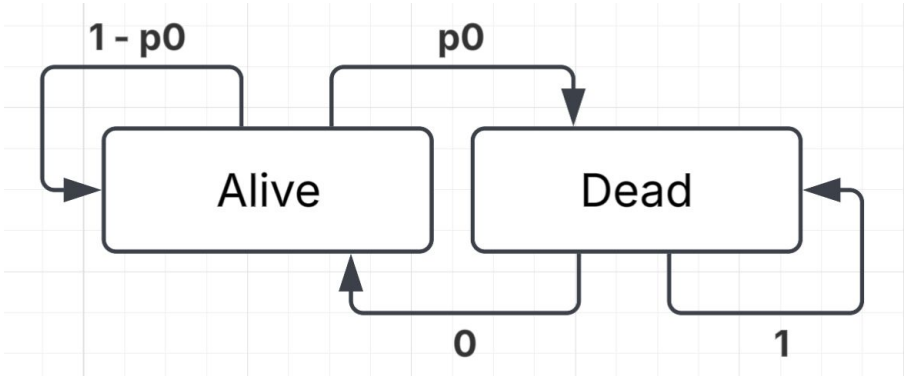
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Markov Chain for Health Transitions

Individuals transition between healthy and disease states over time

p0: probability of disease incidence
p1: probability of recovery





Time Series Data Structure

A tibble: 241,559 × 3 | Groups: hhidpn [38,246]

```
$ ragender     : Factor w/ 2 levels "Female","Male": 2 1 1 1 1 2 2
$ raracem      : Factor w/ 3 levels "White/Caucasian",..: 1 1 1 1
$ agey_e       : num [1:241559] 54 57 59 62 63 56 58 60 62 64 ...
$ raedyrs      : num [1:241559] 16 8 8 8 8 12 12 12 12 12 ...
$ smoken       : num [1:241559] 1 1 1 1 1 0 0 0 0 0 ...
$ lung         : num [1:241559] 0 1 1 1 1 0 0 0 0 0 ...
$ heart        : num [1:241559] 0 0 0 0 0 1 0 1 1 1 ...
$ cancr        : num [1:241559] 0 0 0 0 0 0 0 0 0 0 ...
$ pack_years   : num [1:241559] 47.9 33 33 33 33 ...
$ lung_response: num [1:241559] 0 1 1 1 1 0 0 0 0 0 ...
```

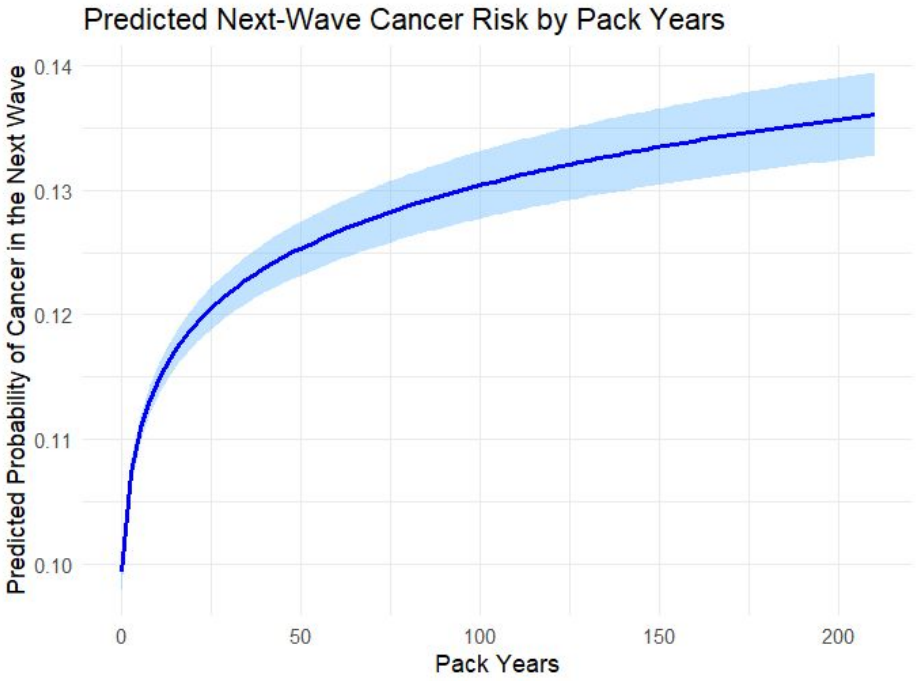
Current lung disease status

Lung disease status in 2 years

hhidpn <dbl>	lung <dbl>	lung_response <dbl>
10003020	0	0
10003020	0	0
10003030	1	1
10003030	1	1
10003030	1	1
10003030	1	0
10003030	0	0
10003030	0	0
10003030	0	1
10003030	1	1

Logistic Regression

- Predictors
  - Demographics
  - Current disease status
  - Log(1 + pack-years)
- Outcomes: Predicting in 2 years
  - Lung disease
  - Heart disease
  - Cancer
  - Death



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Coefficients

Notice that pack-years always has a positive coefficient:
As pack-years increase, so do the odds of getting each disease

Standard Error for pack-years:
Lung: 0.010
Heart: 0.007
Cancer: 0.009
Death: 0.006

Estimated Coefficients

Response is Status in 2 years

	Lung Disease	Heart Disease	Cancer	Death
Intercept	-4.65	-5.18	-5.59	-7.15
Gender: Male	-0.17	0.23	0.30	0.14
Race: Black/AA	-0.08	-0.07	-0.09	0.41
Race: Other	-0.10	-0.14	-0.31	0.47
Age	0.01	0.03	0.02	0.07
Education Years	-0.03	-0.01	0.03	0.00
Current Smoker	0.76	0.08	0.06	0.25
Lung Disease	6.32	0.47	0.13	0.48
Heart Disease	0.47	5.40	0.06	0.39
Cancer	0.30	0.11	7.01	0.49
log(1 + Pack-Years)	0.24	0.05	0.08	0.06

Future Elderly Model: Microsimulation

- Monte Carlo simulation: 10,000 individuals
- Baseline: data from wave 16
- Simulating individual future health outcomes, 15 times (30 years)

Counterfactual Simulation

- A “what-if” analysis, comparing identical populations over the same time
- We change exposure: lifetime smoking
- Our counterfactuals:
  - What if the lifetime pack-years of heavy smokers (>70th percentile) was reduced to 0?
  - What if the lifetime pack-years of light smokers (<30th percentile) was reduced to 0?

Results

04

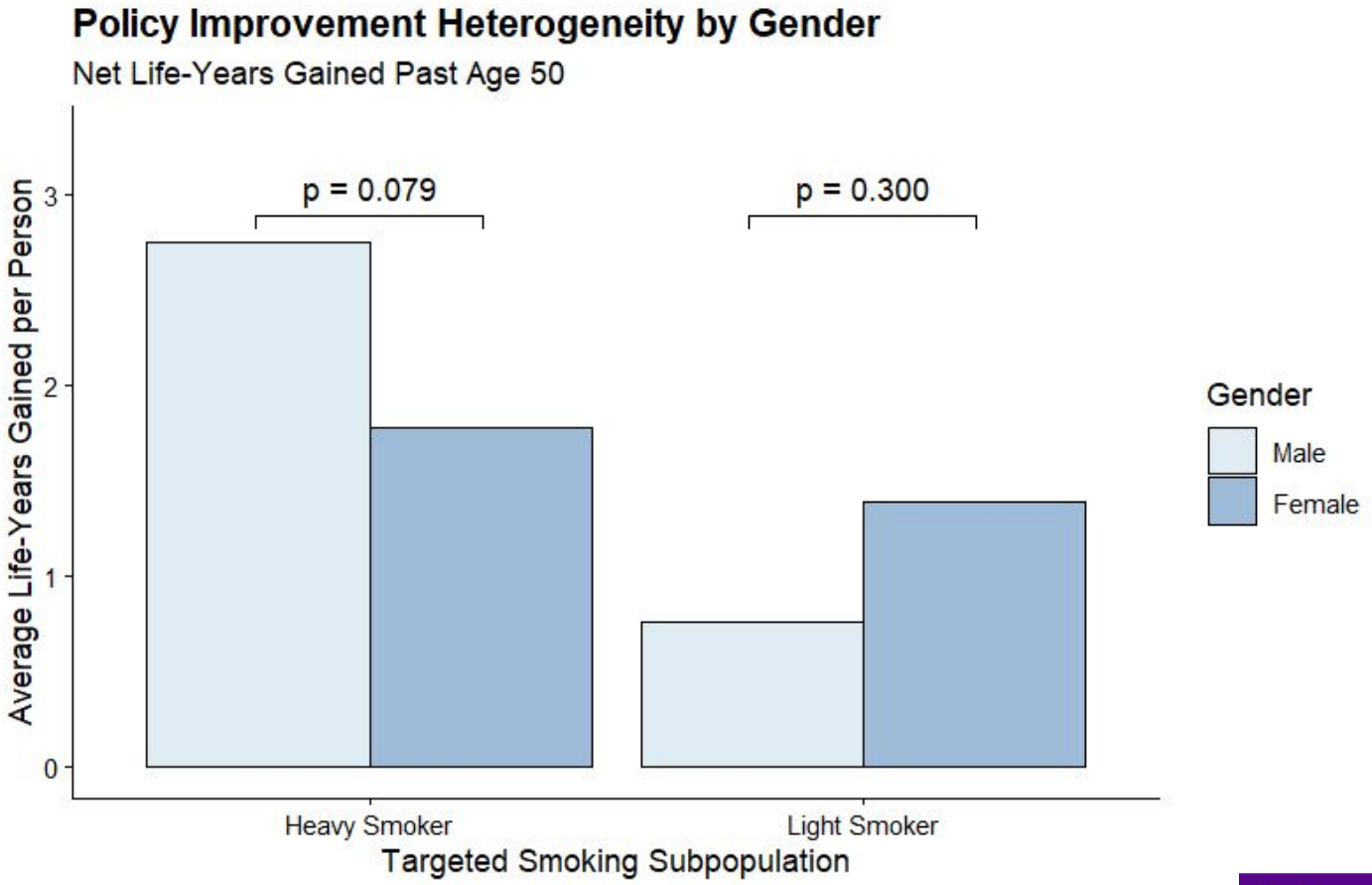
Policies Targeting Light vs. Heavy Smokers

- After heavy smoker cessation: mean life expectancy gain of heavy smokers is **2.23 years**
- After light smoker cessation: mean life expectancy gain of light smokers is **1.09 years**
- The **95% confidence interval** of the true difference between the means is **(0.33, 1.95)**
- The p-value was **0.0056**; the difference is statistically significant

Policy Improvement Heterogeneity by Gender

Gender	Smoking Intensity Tier	Baseline Life Expectancy	Policy Change Life Expectancy	Net Life-Years Gained[1]
Male	Heavy Smoker	59.40	62.15	2.75
Female	Heavy Smoker	60.77	62.55	1.78
Male	Light Smoker	62.88	63.64	0.76
Female	Light Smoker	62.49	63.88	1.39

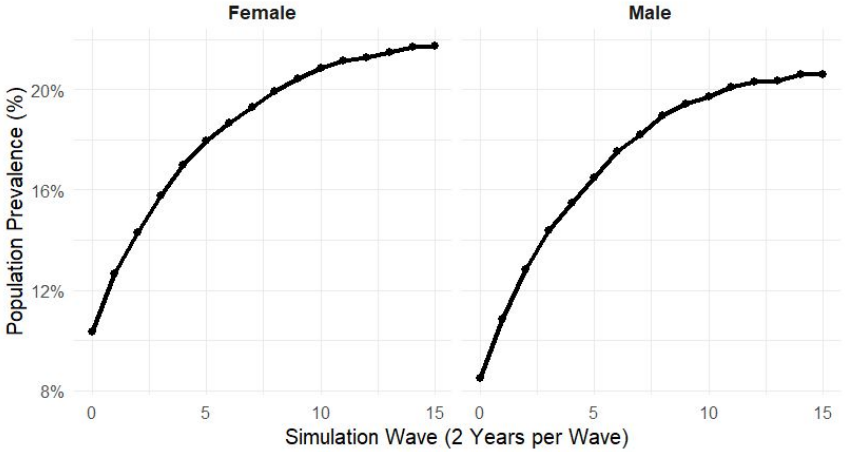
[1] Net Life-Years Gained represents the average individual-level improvement under the specific targeted policy compared to the Status Quo.



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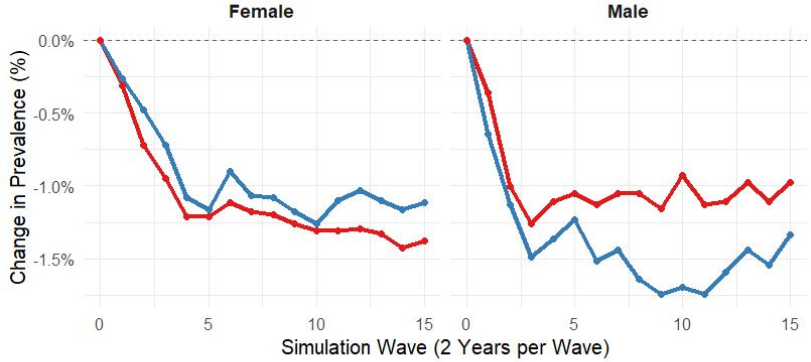
Status Quo Lung Disease Prevalence Trajectory by Gender

Natural progression rate across simulation waves



Net Policy Impact on Lung Disease Prevalence by Gender

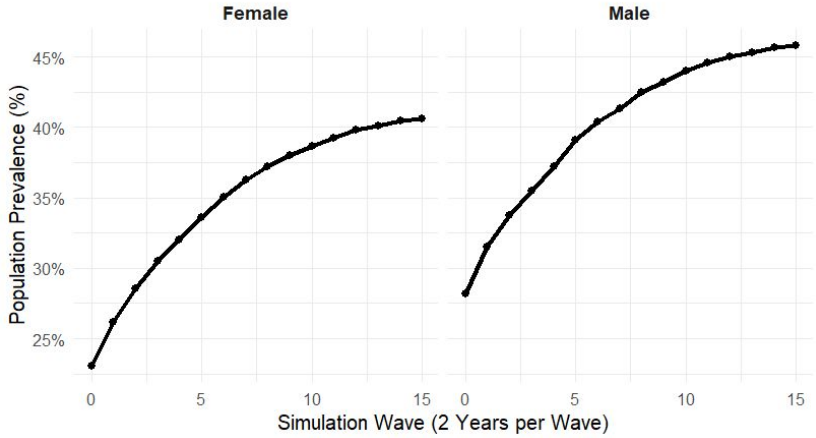
Percentage point change in prevalence compared to the Status Quo



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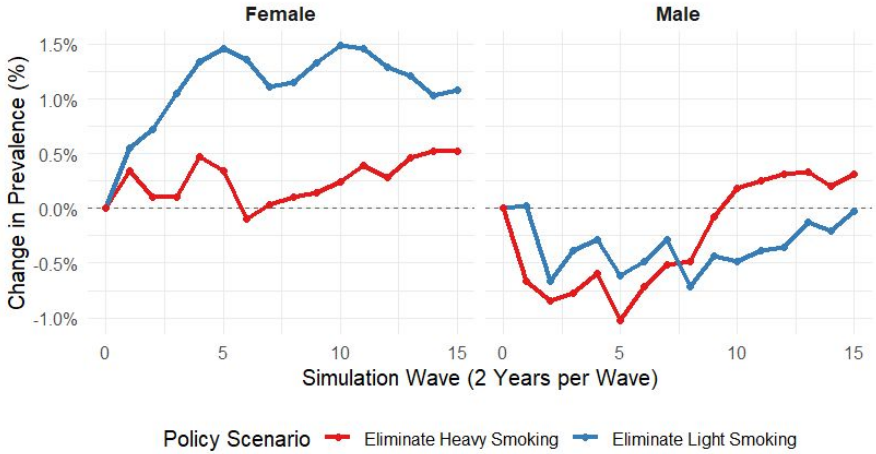
Status Quo Heart Disease Prevalence Trajectory by Gender

Natural progression rate across simulation waves



Net Policy Impact on Heart Disease Prevalence by Gender

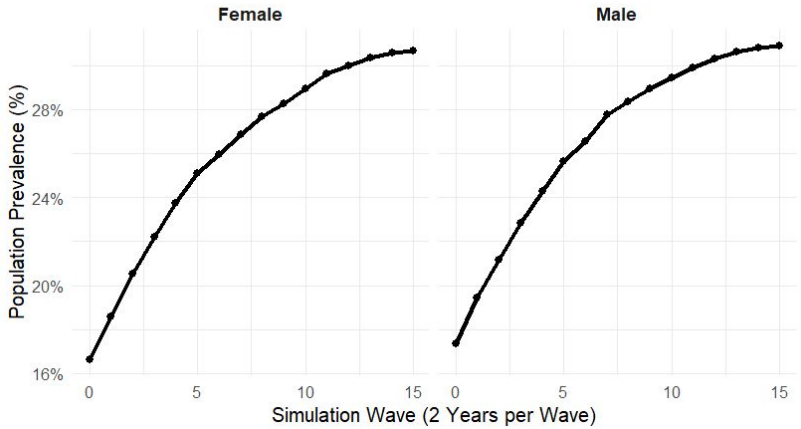
Percentage point change in prevalence compared to the Status Quo



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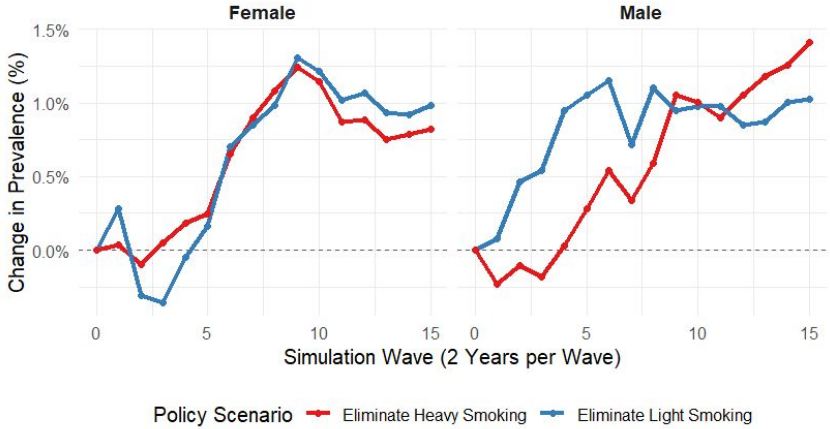
Status Quo Cancer Prevalence Trajectory by Gender

Natural progression rate across simulation waves



Net Policy Impact on Cancer Prevalence by Gender

Percentage point change in prevalence compared to the Status Quo



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Conclusions

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Our Conclusion

- Policies targeting heavy vs. light smokers have significantly different impacts on life expectancy, consistent with prior FEM literature (Kim et al., 2019)
- The same holds for lung disease, while heart disease and cancer are more varied
- The effect of gender was statistically insignificant but still showed a meaningful difference for heavy smokers

Limitations

- We assume consistent lifetime smoking
- Disease categories are broad
- Variation between 2-year waves is not available
- Recall bias, social desirability, survivor bias

Future Directions

- Incorporating longitudinal analysis
  - e.g. accounting for repeated measurements per subject
- Incorporating survival analysis methods
  - e.g. finding age at death
- Extending this model to e-cigarettes
- Using data from real-world smoking cessation policies

Thank you!

Thank you to:

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Our program coordinator, **Vardia Duterville**!

Our program directors, **Dr. Alex Dahlen** and **Dr. Rebecca Betensky**!

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Citations

1. Centers for Disease Control and Prevention. (2015). Current cigarette smoking among adults—United States, 2005–2014. *Morbidity and Mortality Weekly Report, 64*(44), 1233–1240. https://www.cdc.gov/mmwr/preview/mmwrhtml/mm6444a2.htm
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8. WalletHub. (n.d.). *The real cost of smoking by state*. https://wallethub.com/edu/the-financial-cost-of-smoking-by-state/9520
9. Centers for Disease Control and Prevention. (2024). *Cigarette smoking*. U.S. Department of Health and Human Services. https://www.cdc.gov/tobacco/about/index.html

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Thank you!
Questions?

Simulation code excerpt

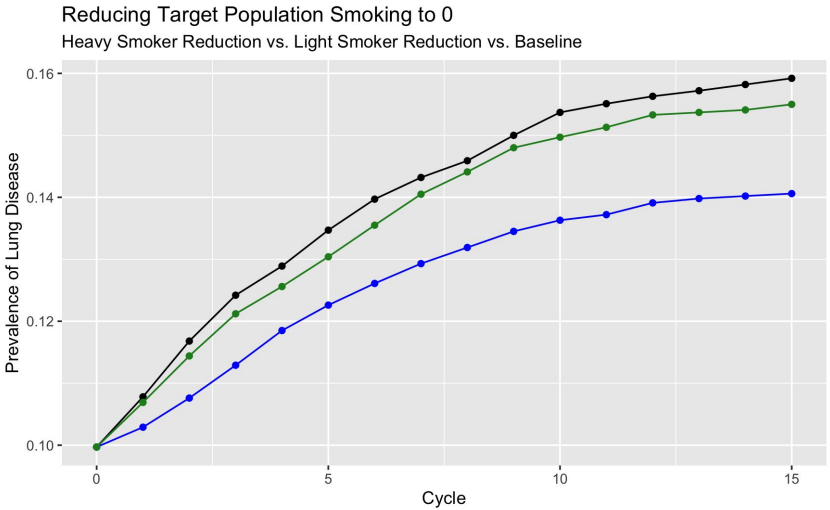
```
  #predicting p0 for each individual based on covariates
  xb_lung <- -4.842251 +
    (-0.179755) * (data$ragender[alive] == "Male") +
    (-0.087264) * (data$raracem[alive] == "African American") +
    (-0.105170) * (data$raracem[alive] == "Other") +
    0.008885  * data$agey_e[alive] +
    (-0.028873) * data$raedyrs[alive] +
    0.734563  * data$smoken[alive] +
    6.592834  * data$lung[alive] +
    0.474956  * data$heart[alive] +
    0.304944  * data$cancr[alive] +
    0.244382  * log(1 + data$pack_years[alive])

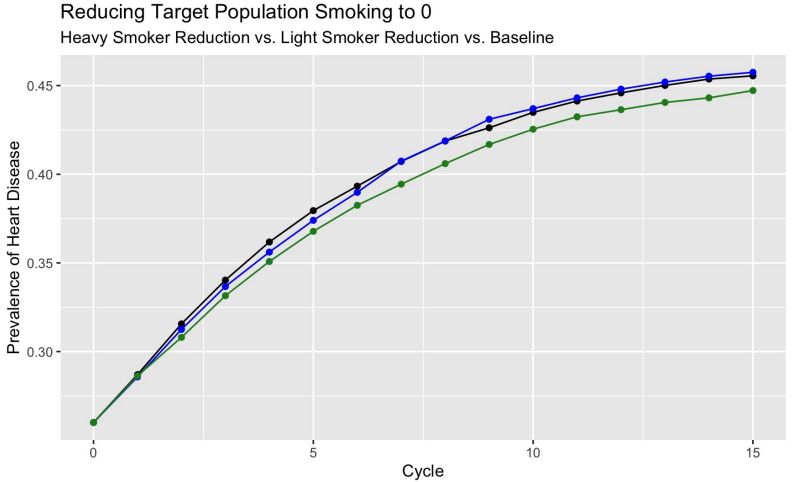
#individual probability for lung disease
  p_lung <- logit_inv(xb_lung)

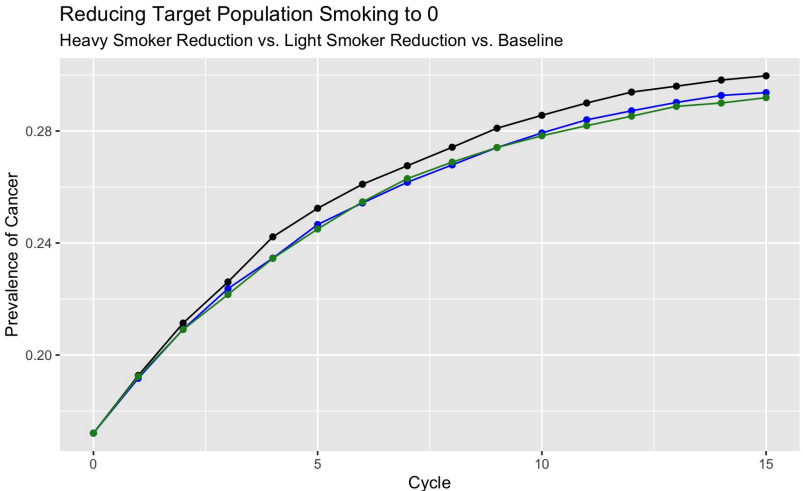
#for each individual, assigning disease status by binomial distribution
data$lung[alive] <- rbinom(length(alive), 1, p_lung)
```

Preparing the Data: Imputation

- Many variables were reported by an individual in one survey wave but missing in others
- Filled constant missing values if reported at least once
    - For example, race or smoking start age
- Missingness in chronic diseases filled similarly
    - We assume chronic disease status is unlikely to change between waves


Reducing Target Population Smoking to 0
Heavy Smoker Reduction vs. Light Smoker Reduction vs. Baseline
Prevalence of Lung Disease
0.16
0.14
0.12
0.10
0
5
10
15
Cycle


Reducing Target Population Smoking to 0
Heavy Smoker Reduction vs. Light Smoker Reduction vs. Baseline
Prevalence of Heart Disease
0.45
0.40
0.35
0.30
0
5
10
15
Cycle


Reducing Target Population Smoking to 0
Heavy Smoker Reduction vs. Light Smoker Reduction vs. Baseline
Prevalence of Cancer
0.28
0.24
0.20
0
5
10
15
Cycle

Difference in Cancer Rates after Heavy Smoker Cessation

